Module 10: Linux Exploitation and Privilege Escalation

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# Exercise 04: Gaining Access and Performing Enumeration using AI
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Scenario

AI-Generated Brute-Force & Automated Post-Exploitation Pipeline leverages Large Language Model (LLM) prompts—such as ChatGPT—to automatically synthesize custom bash scripts combining password auditing tools (**Hydra**) with remote execution and privilege enumeration sequences (**SSH & Linux auditing utilities**).

Key technical concepts in AI-assisted attack automation:
1. **Natural Language Script Synthesis**: Constructing precise prompts requesting a complete attack pipeline:
- Hydra SSH dictionary brute-force (`hydra -l ubuntu -P passwords.txt ssh://192.168.0.51`).
- Parsing cracked credentials into formatted output files (**ssh_credentials.txt**).
- Automated SSH connection establishing initial shell access.
- Remote execution of user, group, and privilege enumeration commands (`cat /etc/passwd` , `groups` , `sudo -l`), exporting results to **user_info.txt**.
2. **Line Ending Sanitization (`dos2unix`)**: Converting Windows-edited or browser-copied scripts containing CRLF line endings to Linux LF format via **dos2unix GainAccess_Enumerate.sh** prior to execution.
3. **Automated Credential Discovery**: Successfully cracking SSH account credentials (**ubuntu** / **toor**) and harvesting system user architecture.

In this exercise, we access ChatGPT, submit a script generation prompt, copy the resulting code into **/home/pentester/Scripts/Module 10/GainAccess_Enumerate.sh**, grant execution permissions (**chmod +x GainAccess_Enumerate.sh**), sanitize format via `dos2unix` , execute the attack pipeline script, and inspect cracked credential file `ssh_credentials.txt` containing password **toor** and enumeration output file `user_info.txt` .

Objective

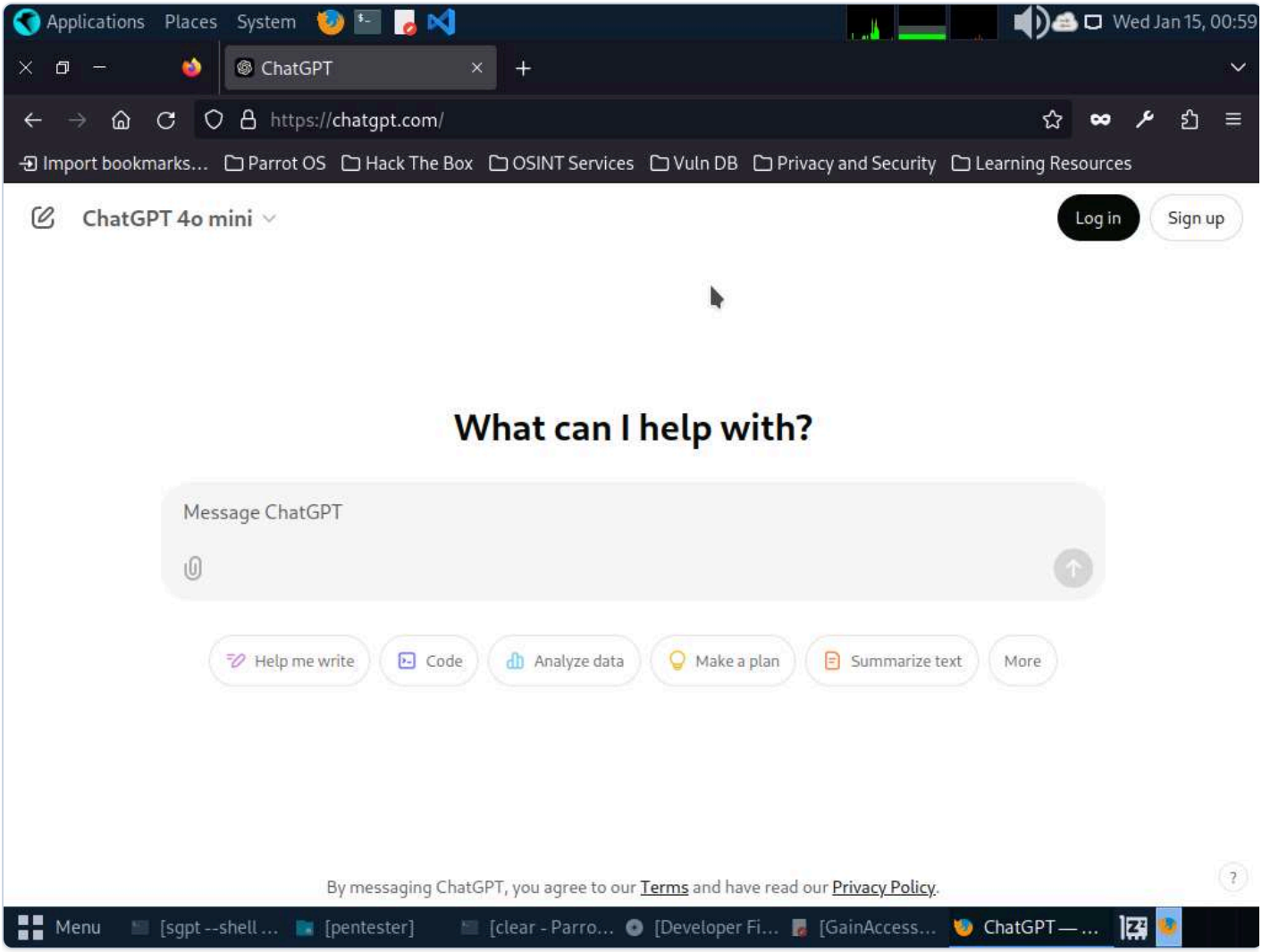
The objective of this lab is to help students learn how to:
- Authenticate to ChatGPT and craft specialized security automation prompts
- Synthesize an integrated Hydra SSH brute-force and remote Linux enumeration script
- Save script to **/home/pentester/Scripts/Module 10/GainAccess_Enumerate.sh**
- Convert DOS line endings to UNIX format: **dos2unix GainAccess_Enumerate.sh**
- Grant execution permissions: **chmod +x GainAccess_Enumerate.sh**
- Execute attack script against target host **192.168.0.51**: **./GainAccess_Enumerate.sh**
- Audit cracked SSH credentials in **ssh_credentials.txt** (Password: **toor**)
- Audit system users and privileges exported in **user_info.txt**

Task 1: Generate AI Shell Script, Sanitize Format via dos2unix, Execute Automated SSH Brute-Force & Enumeration, and Verify Results

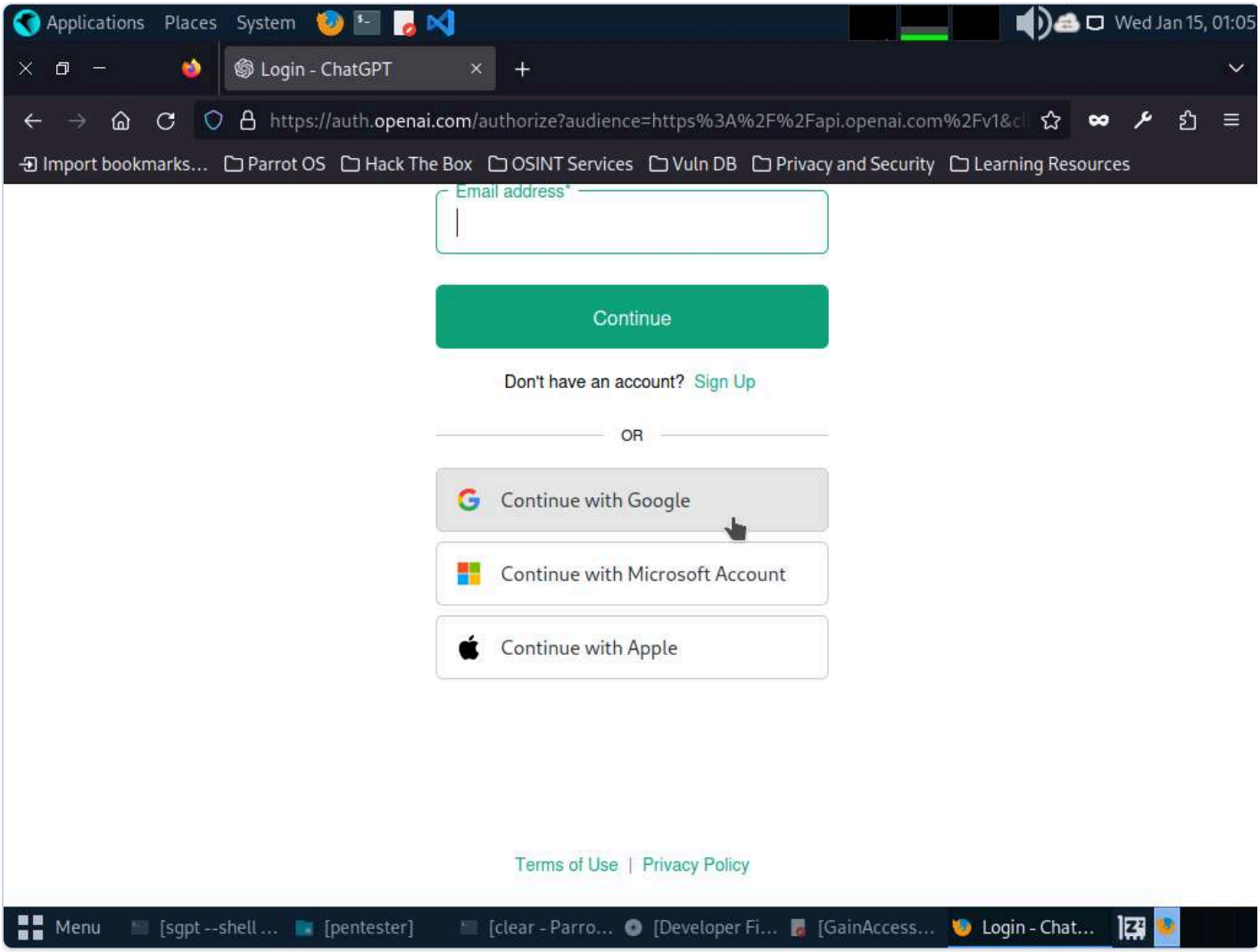
In this task, we prompt ChatGPT to generate `GainAccess_Enumerate.sh` , sanitize formatting, execute Hydra SSH brute-forcing against `192.168.0.51` , crack password `toor` , and audit output files.

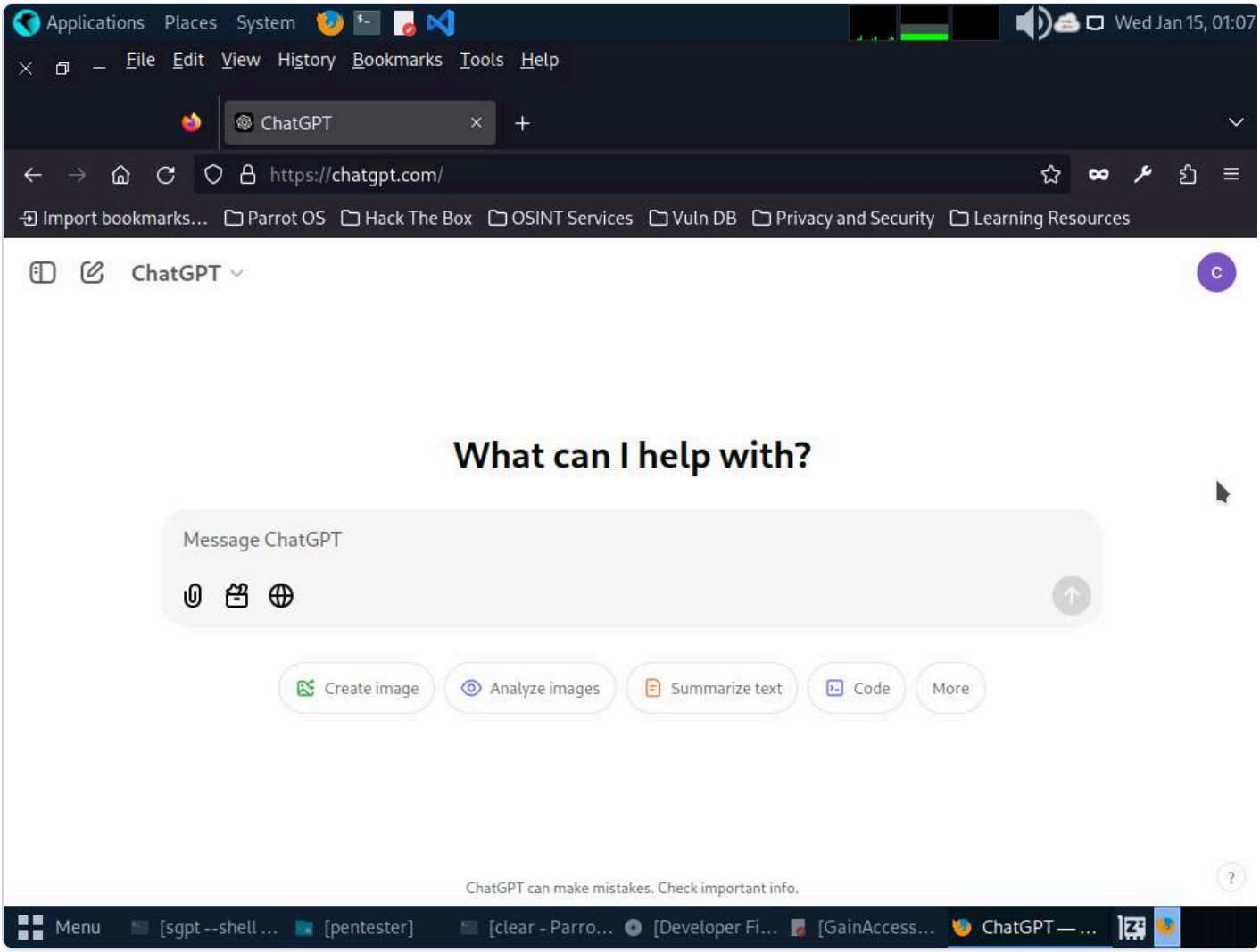
Lab Tasks

1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open Firefox browser and navigate to ChatGPT (`https://chatgpt.com/`).

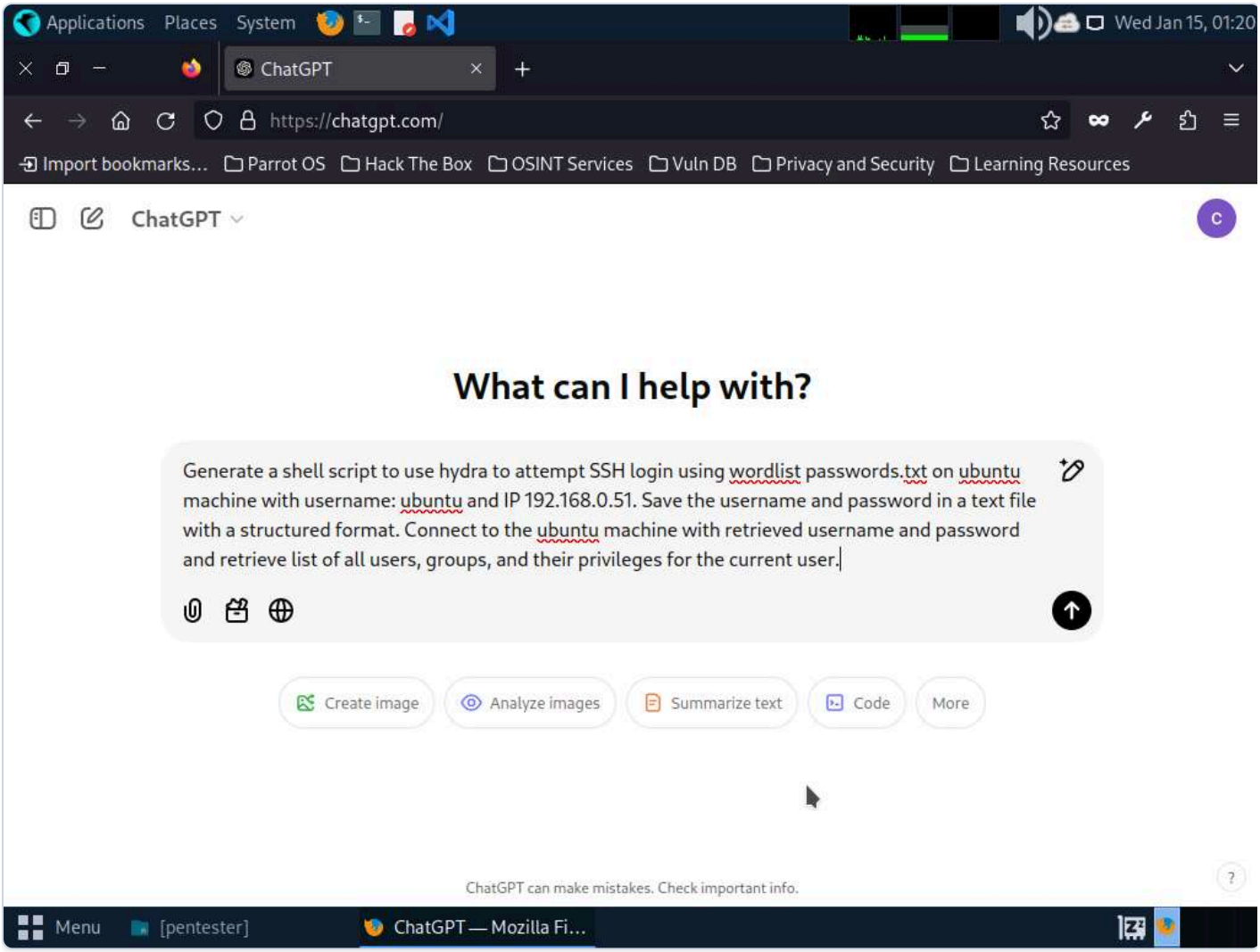


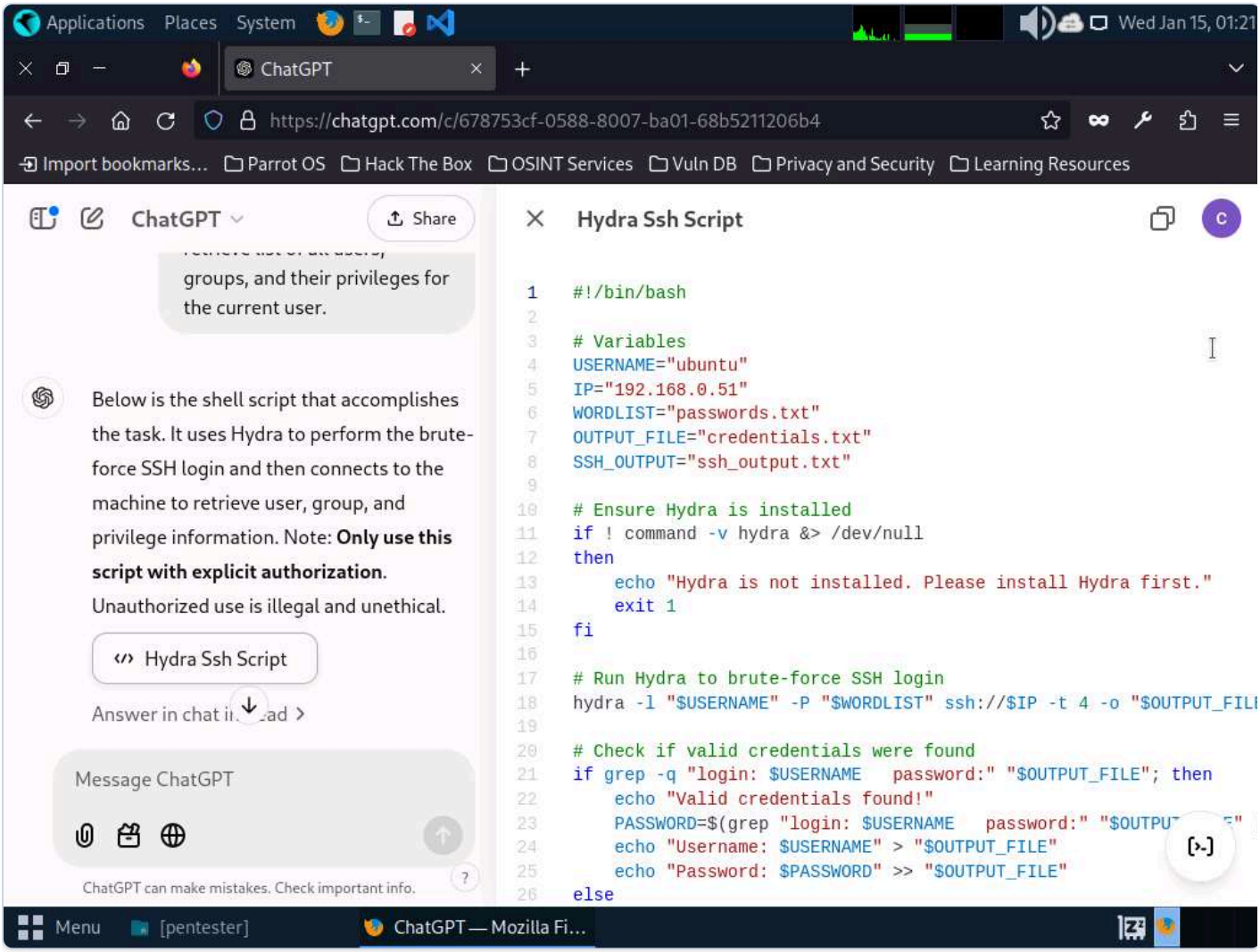
1. Authenticate to ChatGPT using Google credentials.


Applications
Places
System
Wed Jan 15, 01:05
Login - ChatGPT
https://auth.openai.com/authorize?audience=https%3A%2F%2Fapi.openai.com%2Fv1&cl
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
Email address*
Continue
Don't have an account? Sign Up
OR
Continue with Google
Continue with Microsoft Account
Continue with Apple
Terms of Use | Privacy Policy
Menu
[sgpt --shell ...
[pentester]
[clear - Parro...
[Developer Fi...
[GainAccess...
Login - Chat...

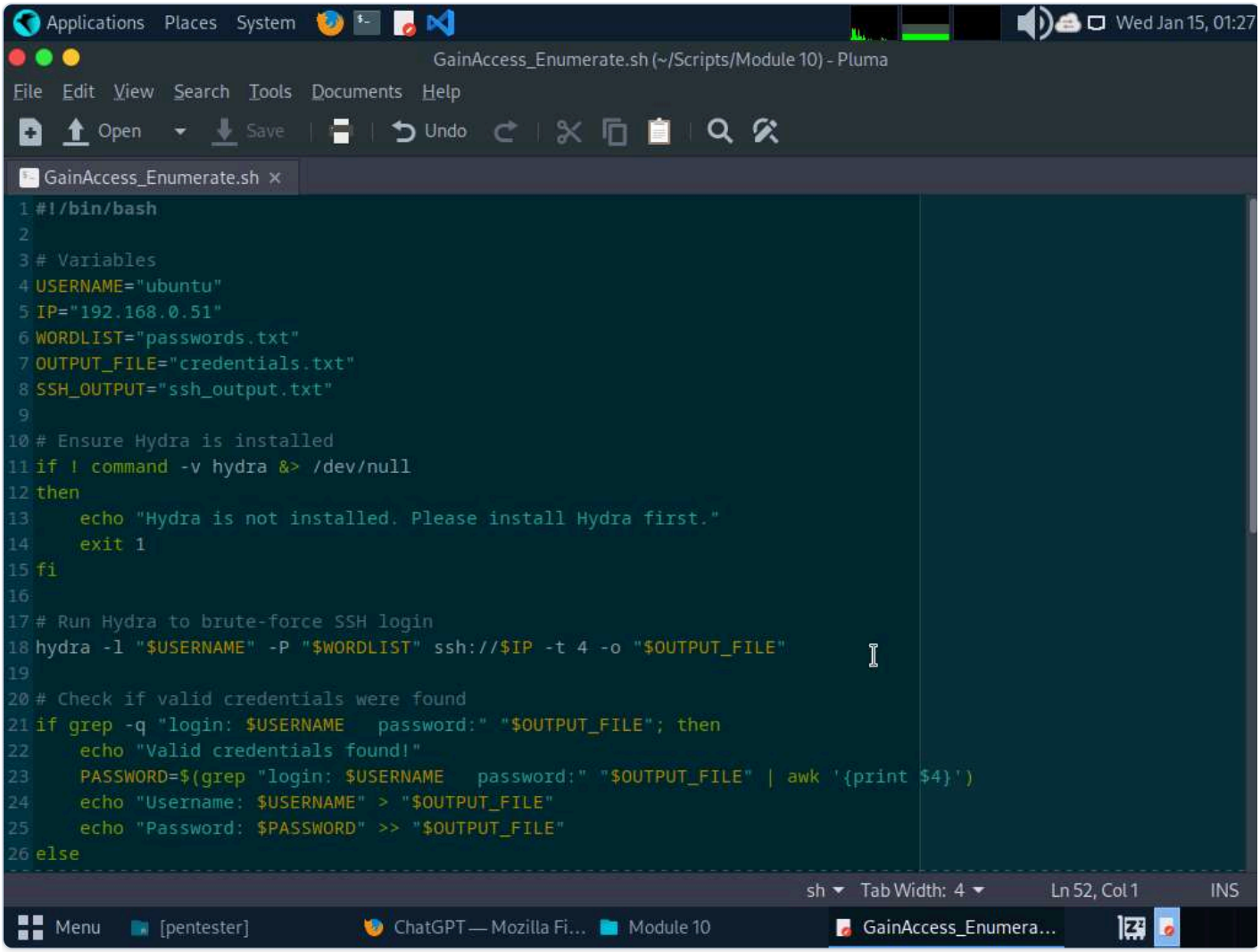


1. Submit the following script generation prompt into ChatGPT:
   Generate a shell script to use hydra to attempt SSH login using wordlist passwords.txt on an Ubuntu machine with username: ubuntu and IP 192.168.0.51. Save the username and password in a text file with a structured format. Connect to the Ubuntu machine with the retrieved username and password and retrieve the list of all users, groups, and their privileges for the current user.

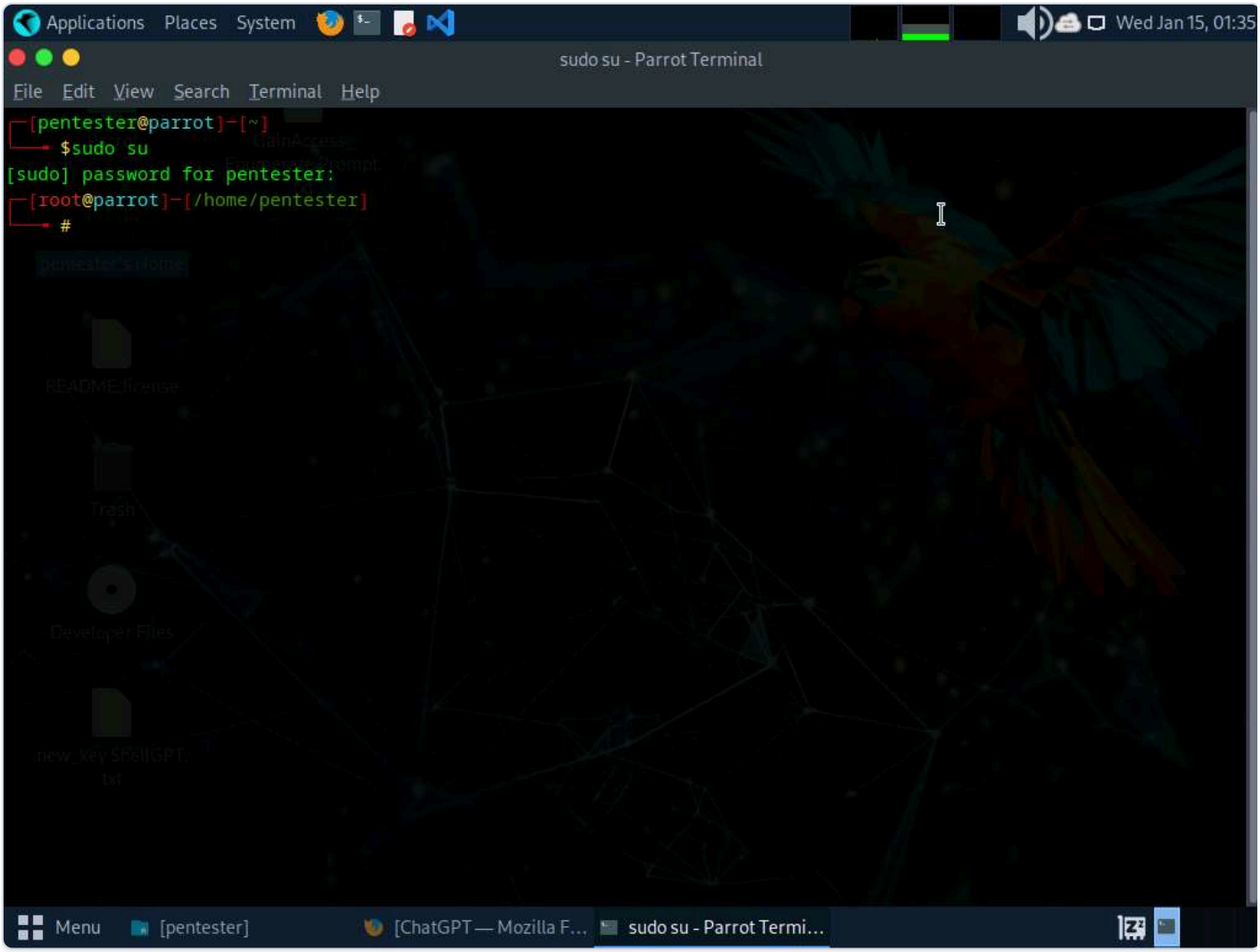

Applications Places System
Wed Jan 15, 01:20
ChatGPT
https://chatgpt.com/
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
ChatGPT
C
What can I help with?
Generate a shell script to use hydra to attempt SSH login using wordlist passwords.txt on ubuntu machine with username: ubuntu and IP 192.168.0.51. Save the username and password in a text file with a structured format. Connect to the ubuntu machine with retrieved username and password and retrieve list of all users, groups, and their privileges for the current user.
Create image
Analyze images
Summarize text
Code
More
ChatGPT can make mistakes. Check important info.
Menu
[pentester]
ChatGPT — Mozilla Fi...



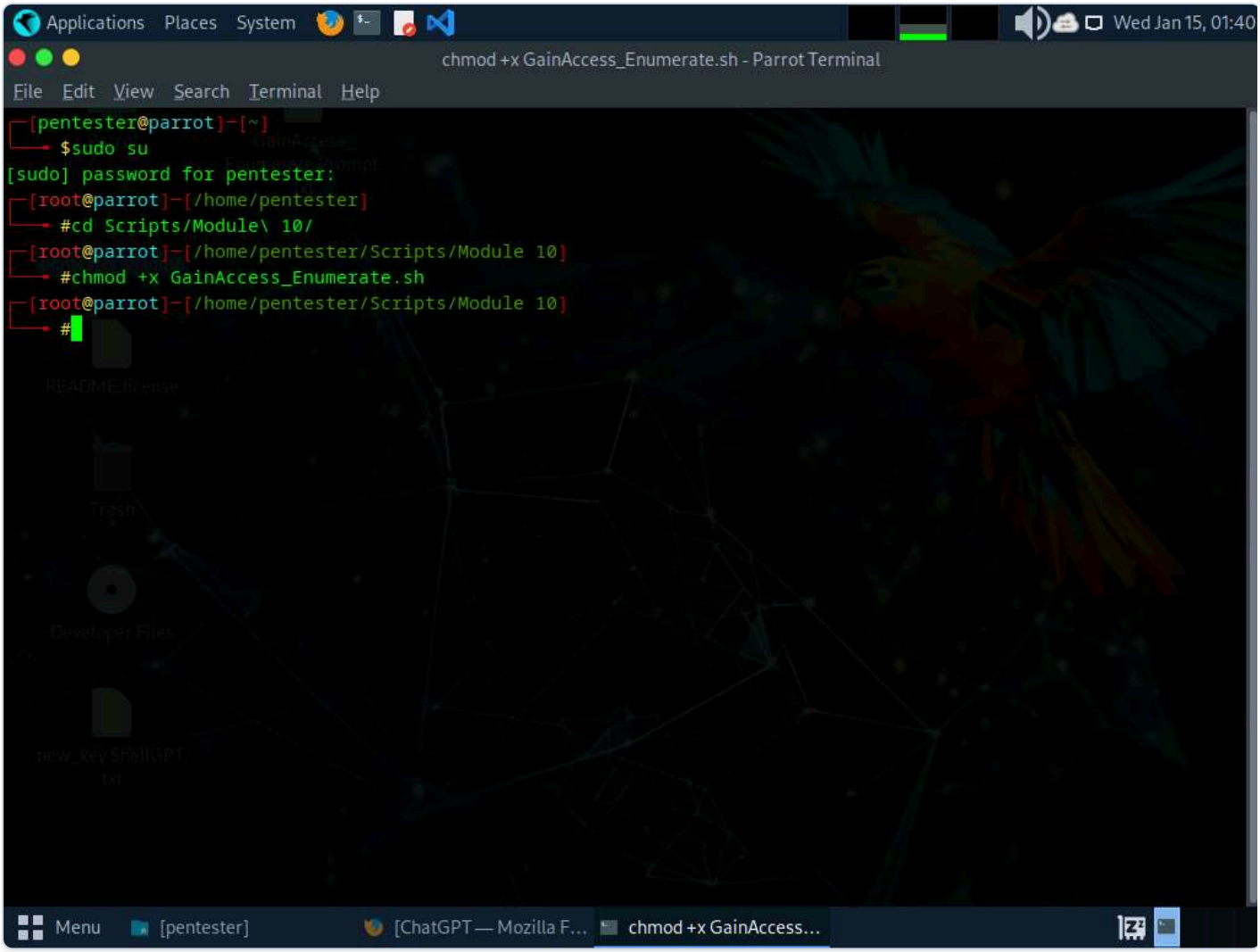
1. Copy the generated shell script code into file **GainAccess_Enumerate.sh**.



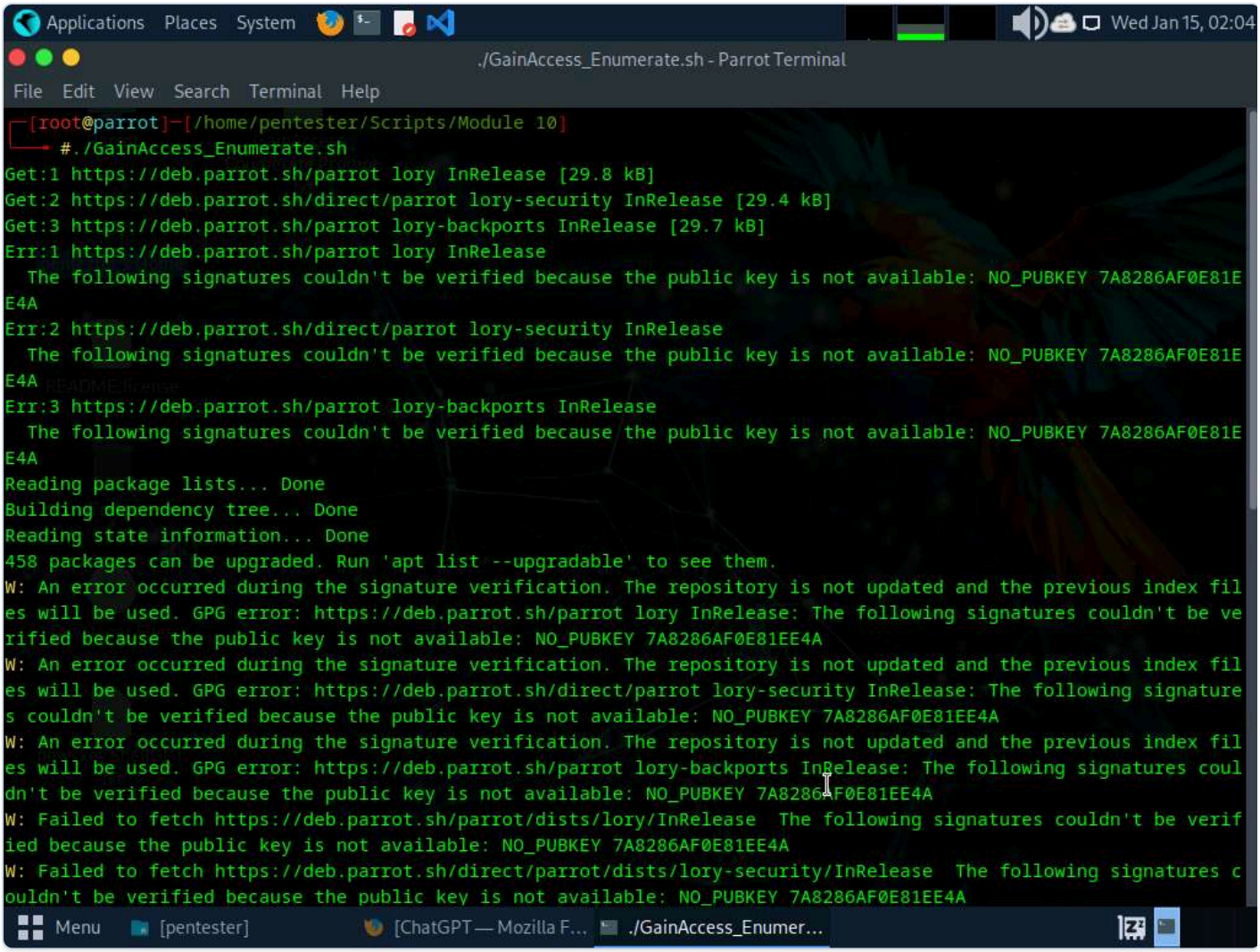
1. Open terminal, obtain root privileges (`sudo su`), and navigate to script directory:
   cd /home/pentester/Scripts/Module\ 10

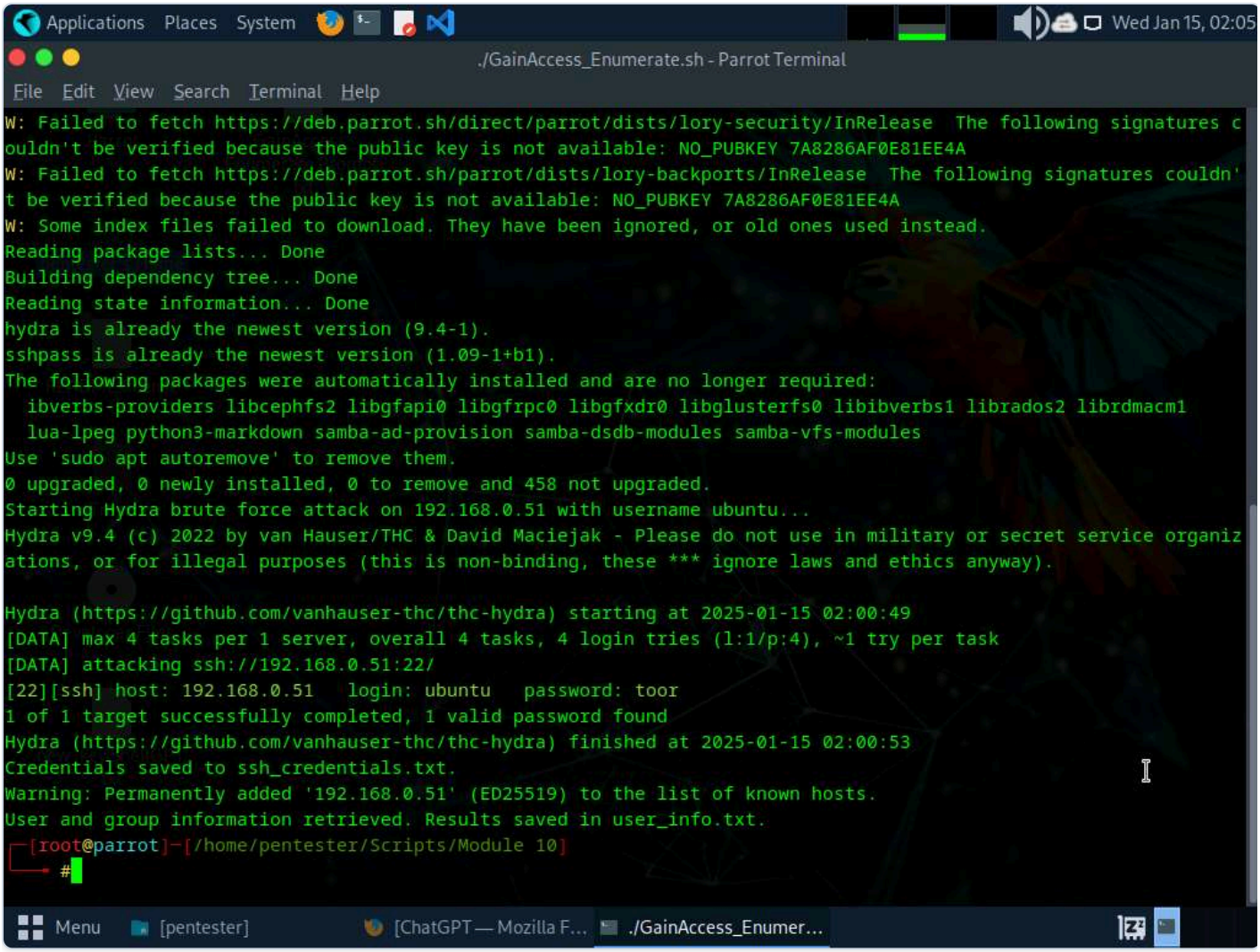


1. Grant executable permissions to script file:
   chmod +x GainAccess_Enumerate.sh



1. Convert script formatting to UNIX line endings and execute attack pipeline:
   dos2unix GainAccess_Enumerate.sh
   ./GainAccess_Enumerate.sh

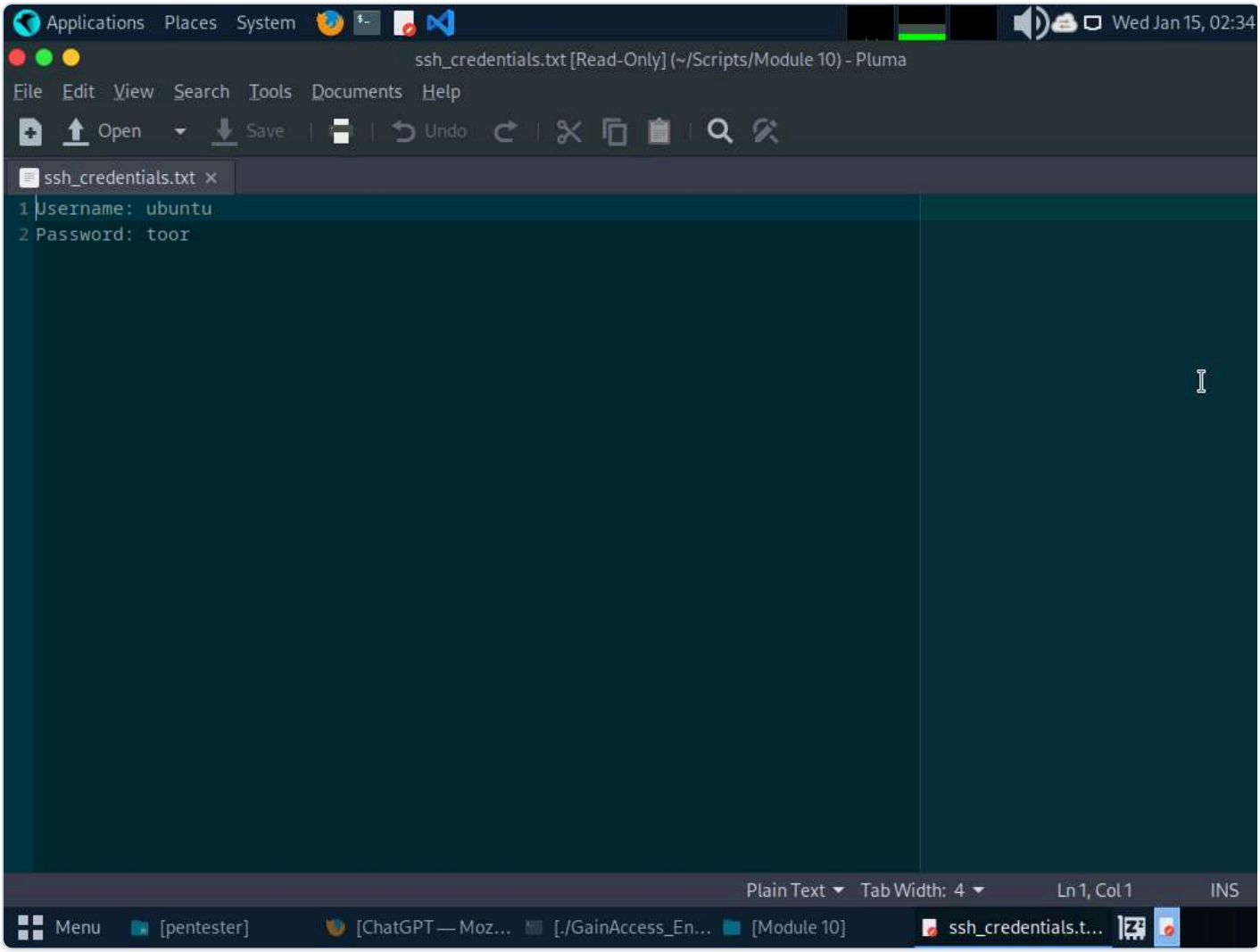

Applications Places System
Wed Jan 15, 02:04
./GainAccess_Enumerate.sh - Parrot Terminal
File Edit View Search Terminal Help
[root@parrot]-[/home/pentester/Scripts/Module 10]
#./GainAccess_Enumerate.sh
Get:1 https://deb.parrot.sh/parrot lory InRelease [29.8 kB]
Get:2 https://deb.parrot.sh/direct/parrot lory-security InRelease [29.4 kB]
Get:3 https://deb.parrot.sh/parrot lory-backports InRelease [29.7 kB]
Err:1 https://deb.parrot.sh/parrot lory InRelease
The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81E
E4A
Err:2 https://deb.parrot.sh/direct/parrot lory-security InRelease
The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81E
E4A
Err:3 https://deb.parrot.sh/parrot lory-backports InRelease
The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81E
E4A
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
458 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: An error occurred during the signature verification. The repository is not updated and the previous index fil
es will be used. GPG error: https://deb.parrot.sh/parrot lory InRelease: The following signatures couldn't be ve
rified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: An error occurred during the signature verification. The repository is not updated and the previous index fil
es will be used. GPG error: https://deb.parrot.sh/direct/parrot lory-security InRelease: The following signature
s couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: An error occurred during the signature verification. The repository is not updated and the previous index fil
es will be used. GPG error: https://deb.parrot.sh/parrot lory-backports InRelease: The following signatures coul
dn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: Failed to fetch https://deb.parrot.sh/parrot/dists/lory/InRelease The following signatures couldn't be verif
ied because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: Failed to fetch https://deb.parrot.sh/direct/parrot/dists/lory-security/InRelease The following signatures c
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./GainAccess_Enumer...



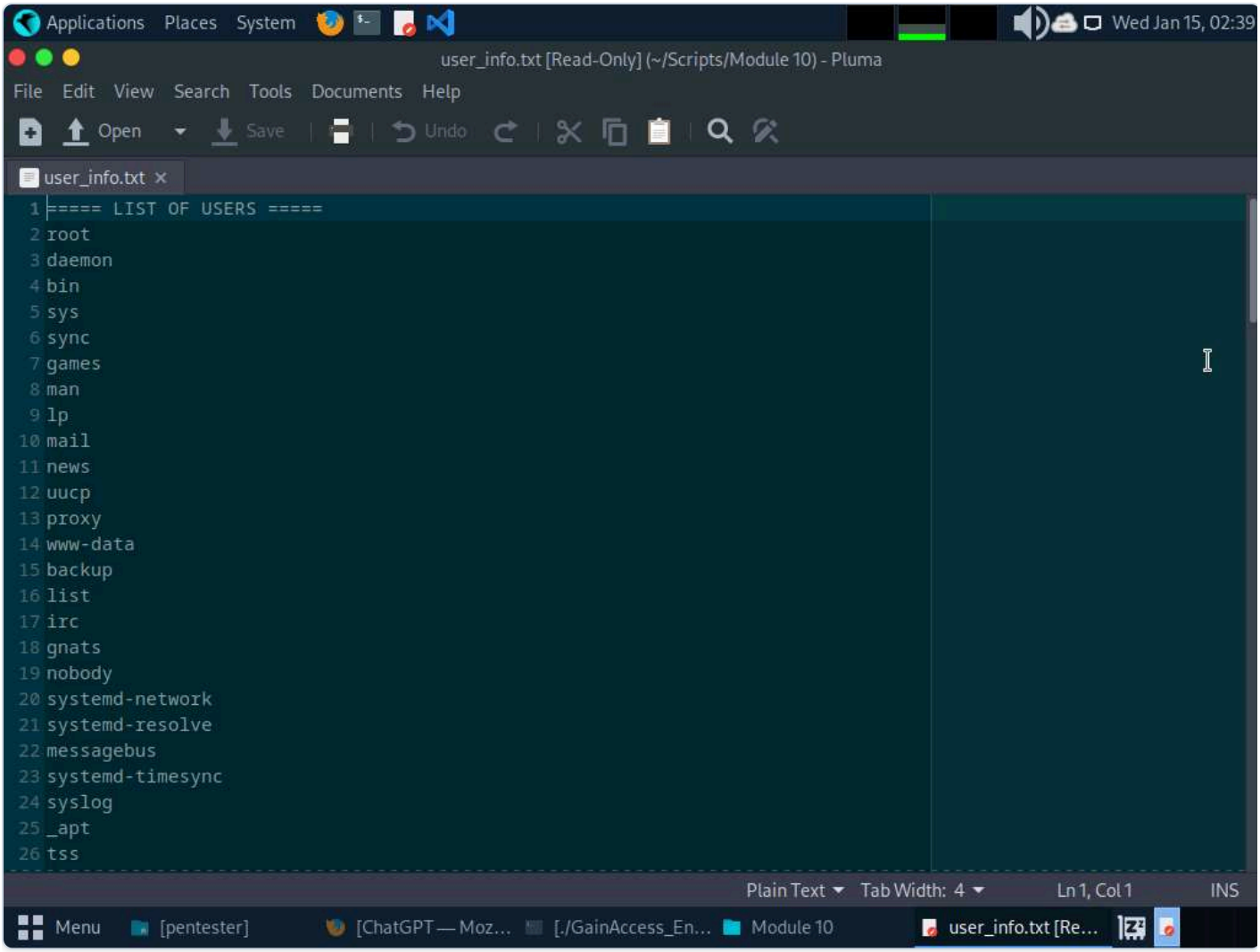
1. Open and inspect cracked credentials file **ssh_credentials.txt**:

   cat ssh_credentials.txt

   Observe output: Hydra successfully cracks SSH password **toor** for user **ubuntu**.



1. Open and inspect enumerated system data file **user_info.txt**:
   cat user_info.txt



This concludes the lab exercise.

IMPORTANT: ❓ **Question 10.4.1.1** On Parrot Security machine, use ChatGPT to generate a script that performs automated ssh brute-force attack on 192.168.0.51 machine to gain access to the machine. Enter the password that was cracked during the ssh brute-force. [!SUCCESS] ✅ **Answer:** **toor**